

SUM Calculation

Data:

Q1 (Ease of Use): Mean = 4.67 (SD = 0.61).

Q2 (Satisfaction): Mean = 4.41 (SD = 0.81).

Q3 (Perceived Time): Mean = 4.33 (SD = 0.83).

The SUM method uses a composite satisfaction score. To obtain it, we calculate the average of the means from the three questions:

- Composite Satisfaction Mean: $(4.67 + 4.41 + 4.33) / 3 = 4.47$.
- Composite Standard Deviation (Estimated): Guilherme, in the paper you do not report the SD of the composite score. What we can do is estimate it using the mean of the individual standard deviations: $(0.61 + 0.81 + 0.83) / 3 \approx 0.75$.
- Task Time Data:
 - Overall Mean Time of 85.31 seconds.
 - STANDARD DEVIATION: the standard deviation of the mean task times is approximately 51.07 seconds.

Standardizations

- Standardized Completion Score
 - Completion Rate: 95.71%.
 - Standardized Score: 95.71
- Standardized Error Score
 - Error Rate: 27.88%.
 - Error-Free Score: $100 - 27.88 = 72.12$
- Standardized Time Score
 - Mean = 85.31 s, Limit = 75 s (assumed), SD = 51.70 s
 - $Z = (75 - 85.31) / 51.70 \approx -0.20$
 - Percentile: 42.1%
- Standardized Satisfaction Score (Composite Mean = 4.47, Composite SD (estimated) = 0.75, Limit = 4.0 (standard for a 5-point scale)).
 - $Z = (4.47 - 4.0) / 0.75 = 0.47 / 0.75 \approx 0.63$
 - Converting the z-score (0.63) to a percentile, we obtain approximately 73.57.
 - Standardized Score: 73.57

PERCENTILE TABLE:

Dimension	Score (%)
Completion	95.71
Errors	72.12
Time	42.10
Satisfaction	73.40

$$\text{SUM} = (95.71 + 72.12 + 42.10 + 73.40) / 4 = 283.33 / 4 \approx 70.83$$

SUM Result

The average of the four standardized scores is calculated:

Our SUM score is 70.83%. This score places EzList usability in the “good” category, because it falls at the threshold of 70, which is considered the start of good usability.

Considerations:

The consolidation of the four performance metrics into the Single Usability Metric (SUM) resulted in a final score of 70.83%. This aggregated indicator places the overall usability of EzList at the threshold of 70, generally considered good. The score was positively influenced by the high task completion rate (standardized score: 95.71) and user satisfaction (standardized score: 73.40), but it was moderated by the error rate (standardized score: 72.12) and, especially, by temporal efficiency (standardized score: 42.10), which stood out as the main area for optimization.

The application of the Single Usability Metric (SUM) made it possible to quantify EzList usability in a single indicator of 70.83%, providing a holistic view that complements the analysis of the individual metrics. Although this score corresponds to a “good” level, it reveals an imbalance: while the platform is highly effective (users are able to complete the tasks) and satisfactory, its efficiency is a point of attention. The low standardized score for task time (42.10) corroborates the findings regarding TBE and the comprehension difficulties, which lead to durations longer than ideal. Therefore, the SUM not only summarizes performance, but also points to the need to focus improvement efforts on optimizing the workflow and clarifying the interface in order to reduce the time and cognitive load required to complete the tasks.